

Decision-Making: Intro to Optimization, Neural Networks, and TinyML

Dr. Erfan Yazdandoost Hamedani
Cody Melcher

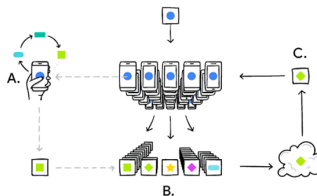
Systems and Industrial Engineering, University of Arizona

Background

- **Dr. Erfan Yazdandoost Hamedani:**

Assistant Professor at UofA

Research: Large-scale Optimization (bilevel, saddle point, distributed)

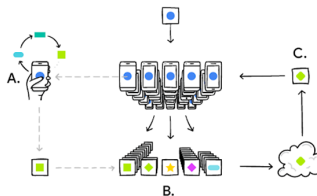


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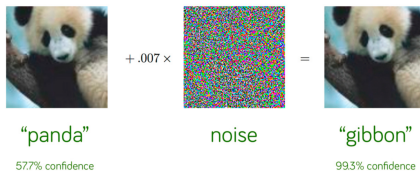
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- **Cody Melcher**

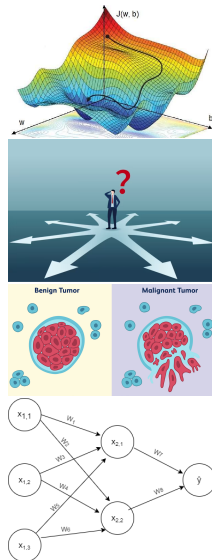
PhD student at UofA

Research: Saddle Point (Min-max) optimization



Overview

- What is optimization?
- How does optimization help with decision-making?
- What is binary classification and how do you solve it?
- What is a neural network?

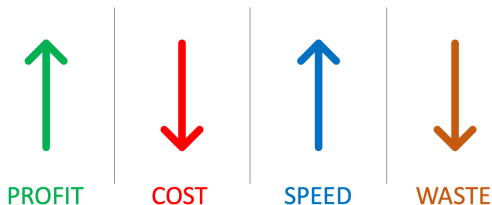


What is Optimization?

- “Optimization” comes from the same root as “optimal”, which means best



- Optimize something means “making it best”

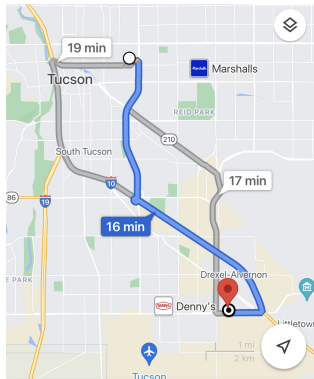


Optimization in real world



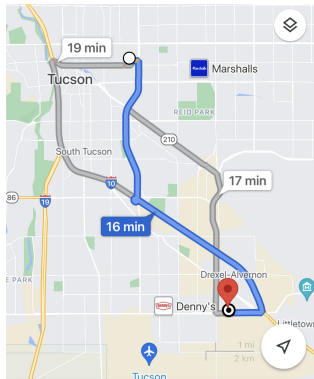
Example

- Going from a school to UofA
- **Goal:** Finding the fastest way
- Different options:
 - Walking
 - Taking Uber
 - Riding bicycle
 - Driving car



Example

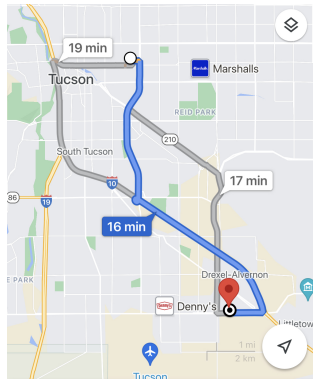
- Going from a school to UofA
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- **Constraint:** Limited budget



Example

- Going from a school to UofA
- **Goal:** Finding the fastest way
- Different options:
 - Walking
 - Taking Uber
 - Riding bicycle
 - Driving car
- **Constraint:** Limited budget
- Optimization problem:

$$\begin{array}{ll} \text{Minimize} & \textit{Time} \\ \text{subject to} & \textit{total cost} \leq \textit{budget} \end{array}$$



Health care – Dental Care

- **Goal:** Minimize cost
- **Variables:** Number of staffs and doctors
- **Constarints:** Patient waiting time should not exceed 30 minutes



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Any uncertainty?

Health care – Dental Care

Uncertainty:

- Service times (duration of surgeries)
- Demand for service (cancellation)
- Availability of resources (equipment malfunctions)



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Research Questions:

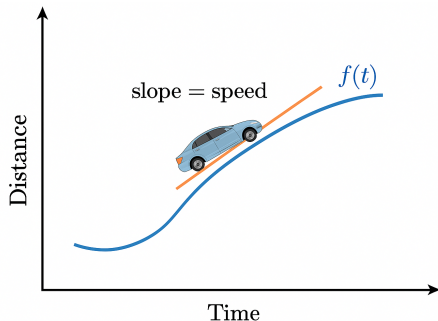
- Modeling the problem mathematically
- Solving the problem
 - Introducing more efficient and accurate algorithms

How to Solve Optimization Problems?

Minimize objective function $f(t)$

Derivative:

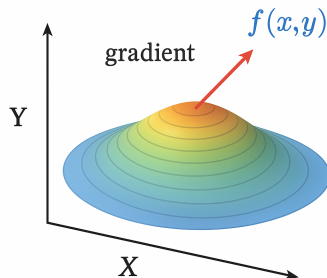
- Measures the rate of change of a function.
- For a curve, the derivative is the **slope** of a tangent line at a point.
- If your function measures distance, then the derivative measures **speed**.



How to Solve Optimization Problems?

Gradient:

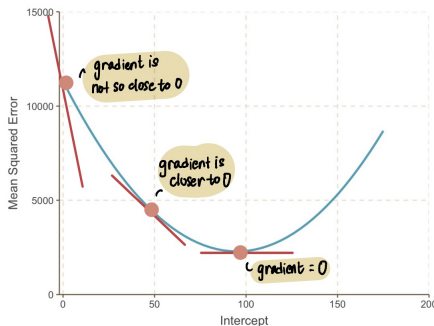
- A gradient is the multi-dimensional version of a derivative.
- It's a **list of derivatives**, one for each direction.
- It points in the **direction of greatest change**



How to Solve Optimization Problems?

Optimization:

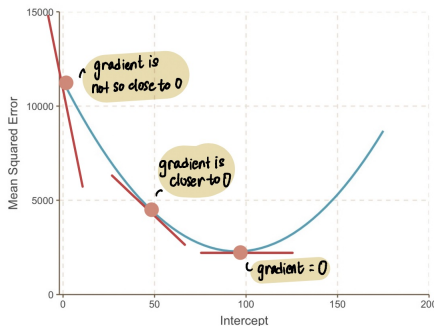
- Optimization means finding the "best" value, frequently the **minimum** of a function.



How to Solve Optimization Problems?

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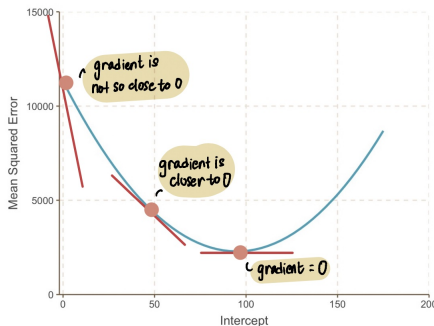
- Optimization means finding the "best" value, frequently the **minimum** of a function.
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How to Solve Optimization Problems?

Optimization:

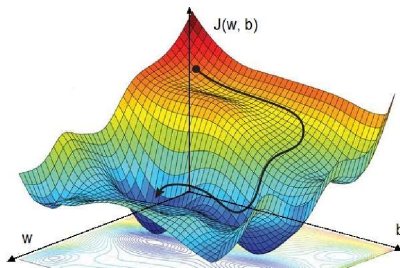
- Optimization means finding the "best" value, frequently the **minimum** of a function.
- The gradient points in the direction where the function increases the fastest. To minimize, we move in the *opposite* direction.
- Visually, it's like finding the bottom of a valley. At the minimum, the gradient is **zero**.



Application: Gradient Descent

Gradient Descent:

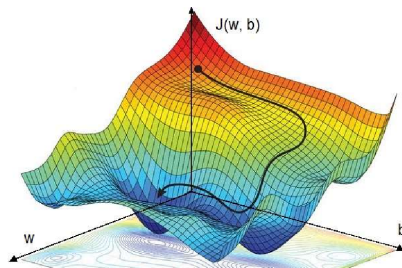
- Imagine standing on a hill and trying to reach the bottom
- Take a step downhill in the direction of steepest slope. Repeat



Application: Gradient Descent

Gradient Descent:

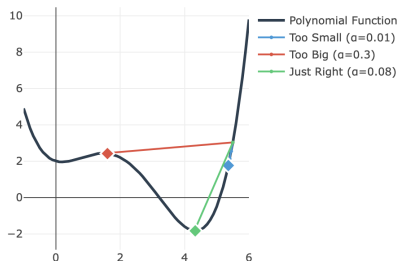
- Imagine standing on a hill and trying to reach the bottom
- Take a step downhill in the direction of steepest slope. Repeat
- Notice that your step size can vary



Step Size Matters!

Why does step size matter?

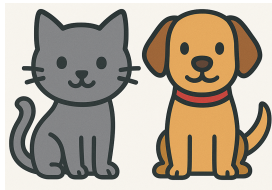
- Gradient descent moves using both the **direction** and the **step size**.
- A good step size leads to fast, stable convergence.
- If it's too small: the model learns very slowly.
- If it's too large: we can overshoot the minimum and bounce back and forth — or even diverge.



*Same starting point — different step sizes,
different outcomes.*

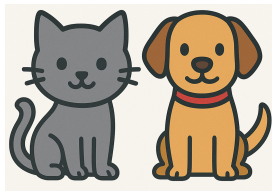
What is Binary Classification?

- Have you ever had to choose between two options (yes or no, left or right, cat or dog)?

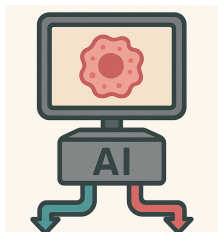


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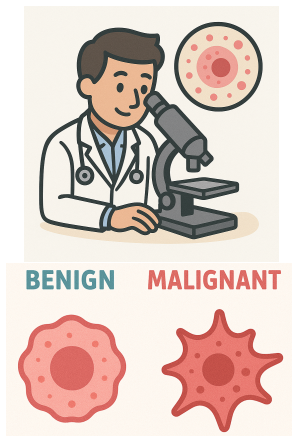


- Binary classification is when a computer learns to decide between **two categories**.
- *Example: Is a cell cancerous or not?*



Healthcare Application

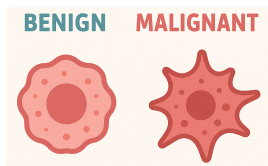
- Doctors take a sample of cells and want to know:
- *Is this tumor **BENIGN** (safe) or **MALIGNANT** (dangerous)?*
- **Goal:** Train a computer to predict that!



What Does the Computer See?

Features of a cell:

Feature	Description	Example
Size	How big the cell is	12.5 μm
Shape	Round or irregular?	Irregular
Texture	Smooth or rough	Rough

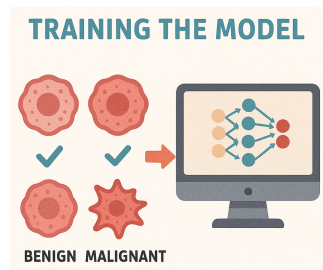


The computer learns from these features to predict whether a new cell is benign or malignant.

Training the Model

How does the model learn?

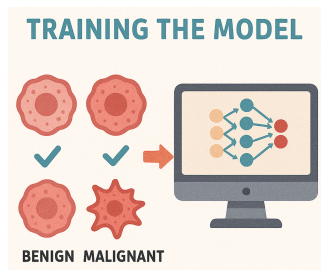
- Doctors label many known cell images as benign or malignant
- The model finds patterns in those labeled examples



Training the Model

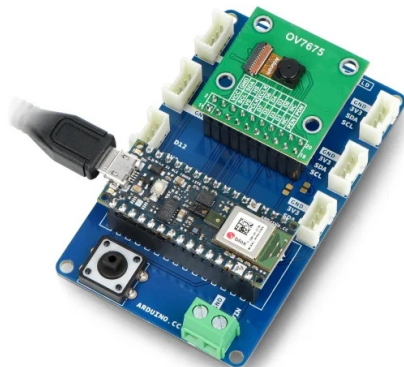
How does the model learn?

- Doctors label many known cell images as benign or malignant
- The model finds patterns in those labeled examples
- The difference is measured using a **loss function**:
 - A number that tells how “wrong” the prediction is
 - **Goal:** minimize this loss



What is an Arduino?

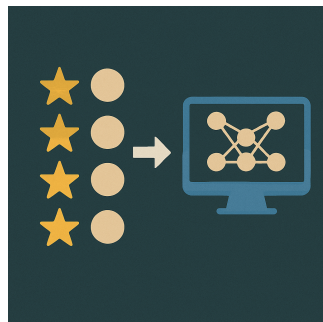
- Arduino is a small computer (microcontroller) built to do specific tasks
- Ours includes a tiny camera and sensors for motion, acceleration, rotation, and sound
- It can run basic machine learning models (TinyML)



Experiment I: Binary Shape Classification

Binary Shape Classification:

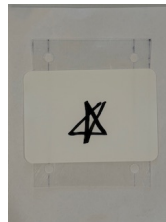
- You will draw a series of stars and circles, show them in turn to the Arduino camera
- These labeled images will be used to train the model
- After 3-5 examples, we'll save the model and test it on an unlabeled star or circle image
- The code uses gradient descent, so feel free to adjust the step size (lr) and see how it changes the result



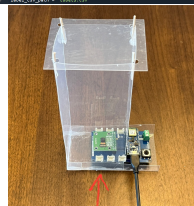
Experiment I – Binary Classification (Part 1)

Step 1: Collect training data

- Draw 3-5 **stars** on sticker cards.
- Place each card on the Arduino camera.
- Run `reading_image.py` and press Enter to save each image.
- Change label (`LABEL=`) in the code to 0, then repeat for 3-5 **circles**.



```
1 import serial
2 import time
3 import os
4
5 # --- Parameters ---
6 PIN = 10
7 BMD = 11200
8 WIDTH_HEIGHT = 176, 144
9 CHOP_0, CHOP_1 = 100, 140
10 CHOP_0, CHOP_1 = (WIDTH - CHOP_0) // 2, (HEIGHT - CHOP_0) // 2
11 DOWNSAMPLED_SIZE = 28
12 BLOCK_0 = CHOP_0 // DOWNSAMPLED_SIZE
13 BLOCK_1 = CHOP_1 // DOWNSAMPLED_SIZE
14 WAIT_TIMES = 1 # How many times to collect
15 LABEL = 0 # 0 for capturing stars, change to 1 if capturing circles
16
17 # --- Output GDI's ---
18 image_save_path = "image_star"
19 label_save_path = "label_star"
```



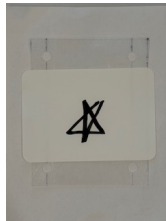
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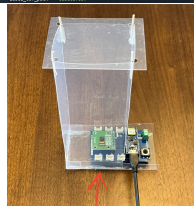
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- Place each card on the Arduino camera.
- Run `reading_image.py` and press Enter to save each image.
- Change label (`LABEL=`) in the code to 0, then repeat for 3-5 **circles**.

Step 2: Train your model

- Run `training_binary.py` to learn from your labeled shapes.
- Try different step sizes (e.g., `lr = 0.1, 1, 10`).
- Click on the plot button to observe how fast or slow the model learns.



```
1 import serial
2 import time as tm
3 import os
4
5 # --- Parameters ---
6 PIN = 10
7 BMD = 11200
8 WIDTH, HEIGHT = 176, 144
9 CHOP_0, CHOP_1 = 100, 140
10 CHOP_2, CHOP_3 = (HEIGHT - CHOP_0) // 2, (HEIGHT - CHOP_0) // 2
11 DOWNSAMPLE_SIZE = 28
12 BLOCK_0 = CHOP_0 // DOWNSAMPLE_SIZE
13 BLOCK_1 = CHOP_1 // DOWNSAMPLE_SIZE
14 HAL_DURES = 1 # How many times to collect
15 LABEL = 0 # 1 for capturing stars, change to 0 if capturing circles
16
17 # --- Output IDs ---
18 IMAGE_CV_PATH = "image.cv"
19 INFO_CV_PATH = "info.cv"
```



Experiment I – Binary Classification (Part 2)

Step 3: Test your model

- Borrow star/circle cards from a neighbor.
- Show each to the Arduino camera.
- Run `testing_binary.py` to predict shape.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # -----
5 # Load Data
6 # -----
7 X = np.loadtxt("images.csv", delimiter=",", dtype=np.float32)
8 y = np.loadtxt("labels.csv", delimiter=",", dtype=np.float32).reshape(-1, 1)
9
10 # Normalize pixels
11 X /= 255.0
12
13 # Convert labels from {0, 1} to {-1, +1}
14 y = 2 * y - 1
15
16 n_samples, n_features = X.shape
17 theta = np.zeros((n_features, 1)) # No bias term
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```

Experiment I – Binary Classification (Part 2)

Step 3: Test your model

- Borrow star/circle cards from a neighbor.
- Show each to the Arduino camera.
- Run `testing.binary.py` to predict shape.

Step 4: Evaluate and Reflect

- How many predictions were correct?
- Were either shape harder to classify?

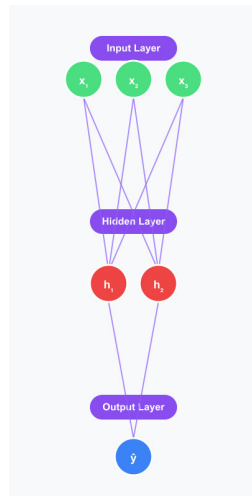
```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Load Data
5
6 # -----
7 X = np.loadtxt('images.csv', delimiter=',', dtype=np.float32)
8 y = np.loadtxt('labels.csv', delimiter=',', dtype=np.float32).reshape(-1, 1)
9
10 # Normalize pixels
11 X /= 255.0
12
13 # Convert labels from {0, 1} to {-1, +1}
14 y = 2 * y - 1
15
16 n_samples, n_features = X.shape
17 theta = np.zeros((n_features, 1)) # No bias term
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```
1 import serial
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 # --- Parameters ---
6 PORT = 'COM1'
7 BAUD = 115200
8 WIDTH, HEIGHT = 176, 144
9 CROP_W, CROP_H = 160, 140
10 CROP_X, CROP_Y = (WIDTH - CROP_W) // 2, (HEIGHT - CROP_H) // 2
11 DOWNSAMPLED_SIZE = 30
12 BLOCK_W = CROP_W // DOWNSAMPLED_SIZE
13 BLOCK_H = CROP_H // DOWNSAMPLED_SIZE
14
15 # --- Load trained weights (theta) ---
16 theta = np.load('theta.npy') # shape: (400,)
17 theta = theta.flatten()
18 assert theta.shape == (400,), "Expected theta to be a (400,) vector"
19
20 # --- Sigmoid Function ---
21 def sigmoid(z):
22     return 1 / (1 + np.exp(-z))
```

What is a Neural Network?

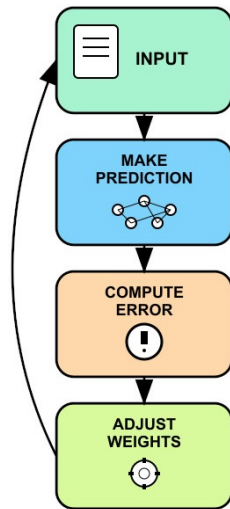
Neural Networks:

- A neural network is a system that learns patterns from data
- It has layers of simple units called **neurons** that do basic math
- Inspired by how the human brain works
- **Feed-forward:** data flows from input to output
- Each layer transforms the input step by step to make a prediction



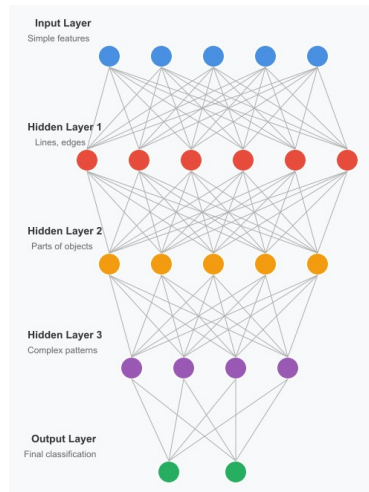
How Does a Neural Network Learn?

- Give the NN an image (input), it then predicts the label of the image
- Compare the predicted label to the actual label (compute the error)
- Use gradient descent to adjust the structure of the NN to reduce the error (adjust weights)
- Given NN a new image and repeat cycle



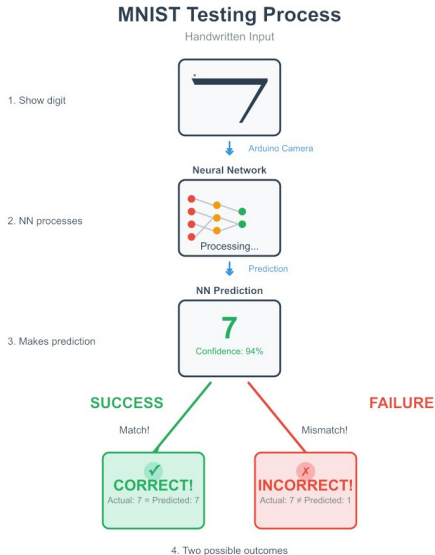
What is Deep Learning?

- A **deep neural network** has many layers stacked together
- Each layer learns something more advanced:
 - First layer: simple things like lines or colors
 - Middle layers: parts of objects (eyes, ears, etc.)
 - Final layers: whole objects (like cats, cars, or numbers)
- It learns step by step by reducing its errors using gradient descent.



Experiment II: Handwritten Number Classification

- We'll use our Arduino kits to classify our handwritten digits via MNIST dataset
- First tweak the given code then hold up a number to the Arduino camera, and it will guess which digit it sees



Experiment II: Arduino Code

Your Task: Modify this code to improve classification

```
model = models.Sequential([
    layers.Input(shape=(28, 28, 1)),
    layers.Flatten(),
    #####
    layers.Dense(100, activation='relu'),
    #layers.Dense(100, activation='relu'),
    #layers.Dense(100, activation='relu'),
    #layers.Dense(100, activation='relu'),
    #####
    layers.Dense(10, activation='softmax')
])
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

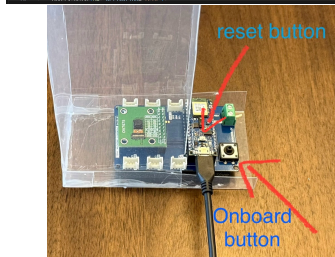
Constraints: Max 4 hidden layers and 900 neurons total • Optimizer choices: Adam/SGD/RMSprop • Activation choices: ReLU/tanh/sigmoid
See handout for details

Experiment II – Digit Classification (TinyML)

Step 1: Upload and Prepare

- Run `training_MNIST.py` to export the model.
- Upload `handwritten_digit_recognition.ino` via Arduino IDE.

```
training_MNIST.py 2 X
Users > codymelcher > Desktop > ArduinoProjects > SEA Arduino-selected > training_MNIST.py ...
1 import tensorflow as tf
2 import numpy as np
3 from tensorflow.keras import layers, models, regularizers
4 from tensorflow.keras.datasets import mnist
5
6 # Load and normalize MNIST data
7 (x_train, y_train), (x_test, y_test) = mnist.load_data()
8 x_train = x_train.astype(np.float32) / 255.0
9 x_test = x_test.astype(np.float32) / 255.0
10 x_train = np.expand_dims(x_train, -1)
11 x_test = np.expand_dims(x_test, -1)
12
13 # L1 regularization strength
14 l1_strength = 3e-5
15
16 # Define a quantization-friendly model
17 model = models.Sequential([
18     layers.Input(shape=(28, 28, 1)),
19     layers.Flatten(),
20     layers.Dense(128, activation='relu', kernel_regularizer=regularizers.L1(l1_strength))
21 ])
```



```
handwritten_digit_recognition_finalized | Arduino IDE 2.3.6
1 #include <TensorFlowLite.h>
2 #include "mnist_model_quant.tfl"
3
4 #include "tensorflow/lite/micro/rtos_realtime.h"
5 #include "tensorflow/lite/micro/micro_interpreter.h"
6 #include "tensorflow/lite/micro/micro_utils.h"
7 #include "tensorflow/lite/micro/micro_error_reporter.h"
8 #include "tinyml.h"
9
10 const int SWIDTH = 176;
11 const int SHHEIGHT = 144;
12 const int OUTWIDTH = 28;
13 const int OUTHEIGHT = 28;
14 const int BLOCKS = SWIDTH * SHHEIGHT; // = 6
15 const int BLOCKS = OUTWIDTH * OUTHEIGHT; // = 3
16 const int CRODIZE = 28;
17
18 Output
```

Experiment II – Digit Classification (TinyML)

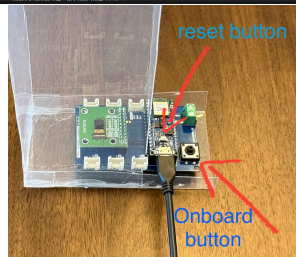
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4 from tensorflow.keras.datasets import mnist
5
6 # Load and normalize MNIST data
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8 x_train = x_train.astype(np.float32) / 255.0
9 x_test = x_test.astype(np.float32) / 255.0
10 x_train = np.expand_dims(x_train, -1)
11 x_test = np.expand_dims(x_test, -1)
12
13 # L1 regularization strength
14 l1_strength = 1e-5
15
16 # Define a quantization-friendly model
17 model = models.Sequential([
18     layers.Input(shape=(28, 28, 1)),
19     layers.Flatten(),
20     layers.Dense(100, activation='relu', kernel_regularizer=regularizers.L1(l1_strength))
21 ])
```

Step 2: Classify Digits

- Draw digits 0–9 on sticker cards. Be sure to center them on the card.
- Place each above the Arduino camera.
- Press the reset button, wait for it capture the image, then hit the onboard button quickly.
- View the result in the serial monitor.



```
handwritten_digit_recognition_finalized | Arduino IDE 2.3.6
Arduino Nano 33 BLE
handwritten_digit_recognition_finalized.ino model_quant.h
1 #include <TensorFlowLite.h>
2 #include "mnist_model_quant.h"
3
4 #include "tensorflow/lite/micro/ai_logger_resolver.h"
5 #include "tensorflow/lite/micro/micro_interpreter.h"
6 #include "tensorflow/lite/micro/micro_utils.h"
7 #include "tensorflow/lite/micro/micro_error_reporter.h"
8 #include <TinyML.h>
9
10 const int SWIDTH = 170;
11 const int SHHEIGHT = 140;
12 const int OUTWIDTH = 28;
13 const int OUTHEIGHT = 28;
14 const int BLOCKS = (width / outwidth) * 5;
15 const int BLOCKS = (height / outheight) * 5;
16 const int CRODIZE = 20;
17
18 Output
```

Experiment II – Digit Classification (TinyML)

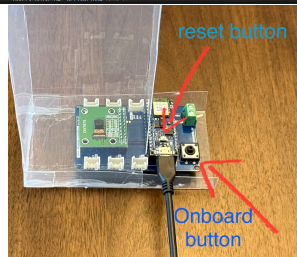
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4 from tensorflow.keras.datasets import mnist
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7 (x_train, y_train), (x_test, y_test) = mnist.load_data()
8 x_train = x_train.astype(np.float32) / 255.0
9 x_test = x_test.astype(np.float32) / 255.0
10 x_train = np.expand_dims(x_train, -1)
11 x_test = np.expand_dims(x_test, -1)
12
13 # L1 regularization strength
14 l1_strength = 1e-5
15
16 # Define a quantization-friendly model
17 model = models.Sequential([
18     layers.Input(shape=(28, 28, 1)),
19     layers.Flatten(),
20     layers.Dense(100, activation='relu', kernel_regularizer=regularizers.L1(l1_strength))
21 ])
```

Step 2: Classify Digits

- Draw digits 0–9 on sticker cards. Be sure to center them on the card.
- Place each above the Arduino camera.
- Press the reset button, wait for it capture the image, then hit the onboard button quickly.
- View the result in the serial monitor.



Ideas

- Modify layers, neurons, or optimizer.
- See how these changes affect accuracy.

```
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Arduino Nano 33 BLE
handwritten_digit_recognition_finalized.ino model_quant.h
1 #include <TensorFlowLite.h>
2 #include "mnist_model_quant.h"
3
4 #include "tensorflow/lite/micro/aiops_resolver.h"
5 #include "tensorflow/lite/micro/micro_interpreter.h"
6 #include "tensorflow/lite/micro/micro_mutable_data_provider.h"
7 #include "tensorflow/lite/micro/micro_error_reporter.h"
8 #include <TinyML.h>
9
10 const int SWITCH = 17;
11 const int SWHEIGHT = 144;
12 const int SWWIDTH = 26;
13 const int SWHEIGHT2 = 26;
14 const int BLOCK = (SWHEIGHT * SWWIDTH) // 4
15 const int BLOCK2 = (SWHEIGHT2 * SWWIDTH2) // 4
16 const int CRODZISE = 28;
17
18 Output
```

Experiment II – Final Testing and Submission

Step 3: Evaluation

- Once you're happy with your model, come to the front table.
- You'll receive a set of **8 pre-printed digit cards** (0–9).
- Test your model on each card using the Arduino camera.
- Record the predicted digit for each one.

Step 4: Report Your Accuracy

- How many digits did your model classify correctly out of 10? Report your accuracy

Things to think about

- Which digits were misclassified?
- Did your model struggle with certain digits?